

Problem 2.17

(a) For the accumulation function $a(t) = (1 + i)^t$, show that

$$\frac{a'(t)}{a(t)} = \ln(1 + i).$$

(b) For the accumulation function $a(t) = 1 + it$, show that

$$\frac{a'(t)}{a(t)} = \frac{i}{1 + it}.$$

Solution

(a) Since

$$a(t) = (1 + i)^t,$$

we justify first that, for a constant $b > 0$,

$$\frac{d}{dt}b^t = b^t \ln(b).$$

Indeed,

$$\frac{d}{dt}b^t = \lim_{h \rightarrow 0} \frac{b^{t+h} - b^t}{h}.$$

Since $b^{t+h} = b^t b^h$, this becomes

$$\frac{d}{dt}b^t = \lim_{h \rightarrow 0} \frac{b^t b^h - b^t}{h} = b^t \lim_{h \rightarrow 0} \frac{b^h - 1}{h}.$$

Now write

$$b^h = e^{h \ln b}.$$

Then, as $h \rightarrow 0$, using $e^x \approx 1 + x$ for small x ,

$$e^{h \ln b} - 1 \approx h \ln b,$$

so

$$\lim_{h \rightarrow 0} \frac{b^h - 1}{h} = \lim_{h \rightarrow 0} \frac{e^{h \ln b} - 1}{h} = \ln b.$$

Therefore,

$$\frac{d}{dt}b^t = b^t \ln b.$$

Applying this with $b = 1 + i$, we obtain Therefore,

$$a'(t) = (1 + i)^t \ln(1 + i).$$

Dividing by $a(t) = (1 + i)^t$, we get

$$\frac{a'(t)}{a(t)} = \frac{(1 + i)^t \ln(1 + i)}{(1 + i)^t} = \ln(1 + i).$$

(b) Since

$$a(t) = 1 + it,$$

its derivative is

$$a'(t) = i.$$

Hence,

$$\frac{a'(t)}{a(t)} = \frac{i}{1 + it}.$$